

A Tale of Five Decoders.

David Ponarovsky

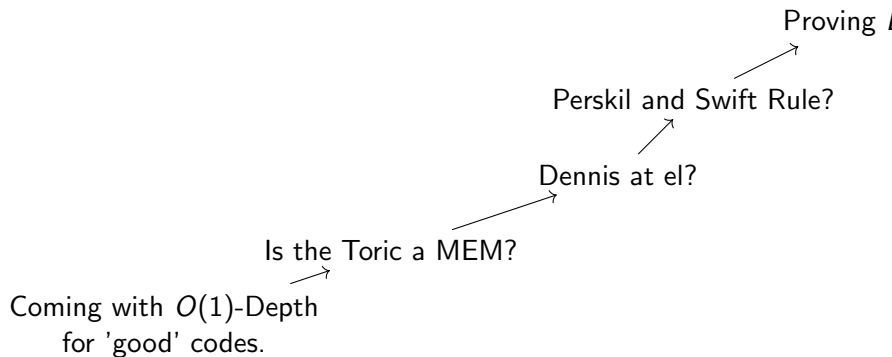
November 2, 2025

Introduction

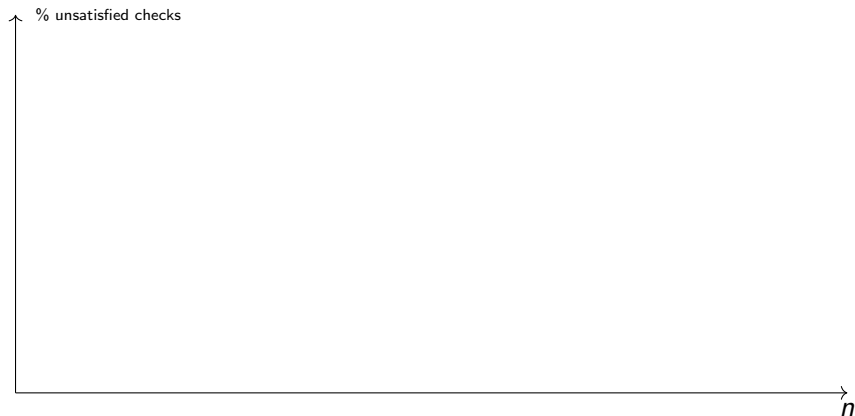
Today:

- ▶ Recap, and where we stood before the summer.
- ▶ Simulations Results.
- ▶ Where we heading.

Map

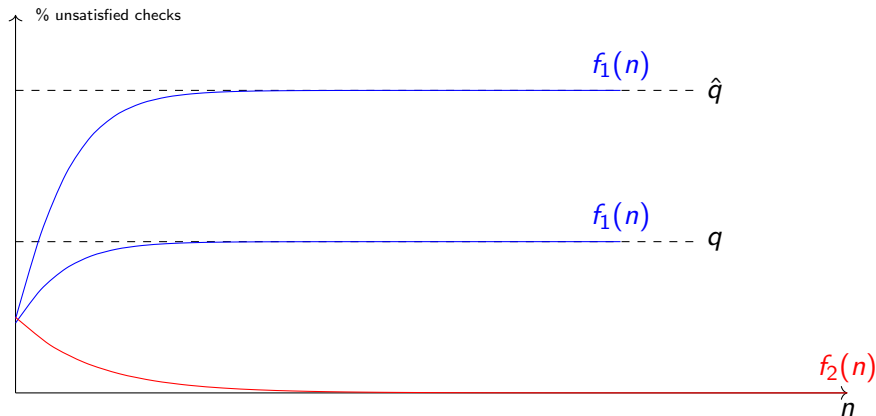


How to Read



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

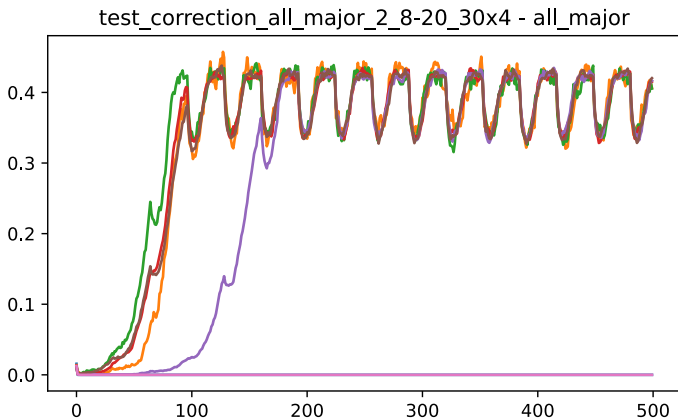
Ideal Result.



colors

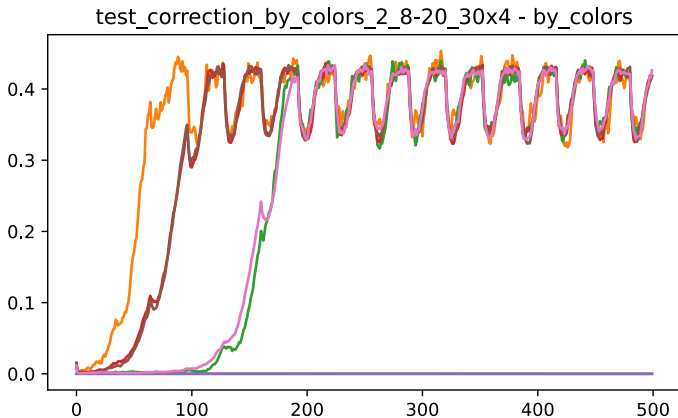
In red, No accumulation of errors. In blue decoding in the presence of noise.

Majority



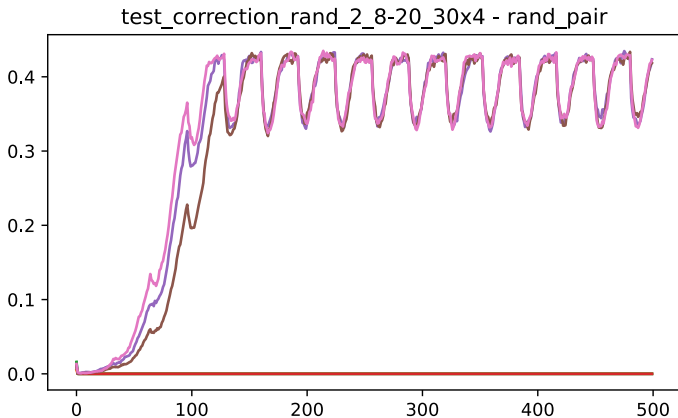
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Colors



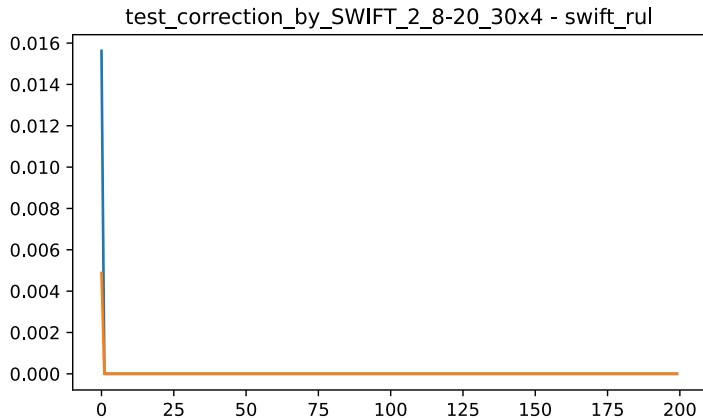
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Picking Random



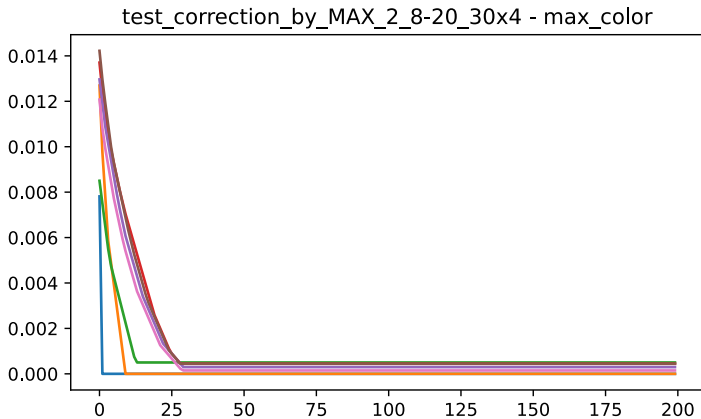
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

SWIFT



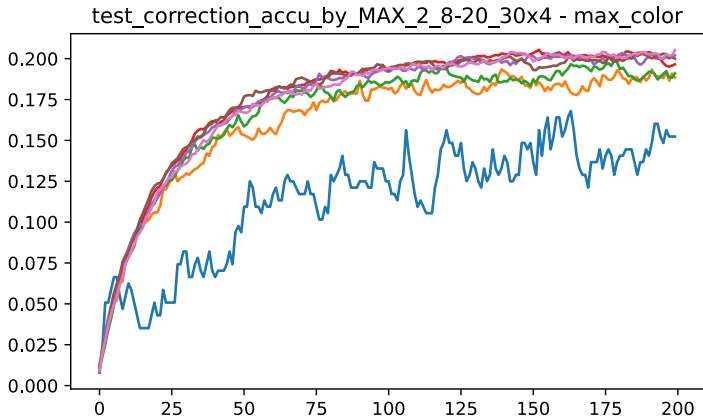
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True